

Dr. Doc:

Upper Respiratory Tract Infections (URTI)

✦ General Overview

▼ Upper Respiratory Tract Infections (URTIs) are infections involving the nasal cavity, pharynx, and tonsils. They are named according to the anatomical site affected. Because neighboring structures are closely connected, an infection can spread from one site to another and produce a mixed infection.

✦ Anatomical Classification

▼ Rhinitis (Coryza): Inflammation of the nasal mucosa. It is the typical manifestation of the common cold and causes increased nasal secretions and obstruction.

▼ Pharyngitis: Inflammation or infection of the pharynx, producing sore throat and painful swallowing. It may be caused by viruses or bacteria.

▼ Tonsillitis: Inflammation of the palatine tonsils, often occurring together with pharyngitis. The tonsils may become red, swollen, and covered with exudate in some bacterial infections.

▼ Mixed Infections: Several adjacent structures may be affected simultaneously. Examples include rhinopharyngitis and tonsillopharyngitis.

✦ Clinical Features of Rhinitis

▼ Rhinitis is the most common presentation of the common cold. The severity varies depending on the causative organism and the individual's immune response.

🎯 Inflammatory Changes: Infection causes inflammation and edema of the nasal mucosa, which narrows the nasal passages and produces nasal obstruction.

▼ Mucus Production: Increased secretion from nasal glands produces rhinorrhea. The discharge is usually clear and watery initially.

▼ As inflammation continues, the discharge may become thicker and purulent over approximately 5–10 days. This change does not necessarily indicate a bacterial infection because inflammatory cells and cellular debris can alter the appearance of mucus.

▼ Fever may occur but is usually variable and mild, particularly in uncomplicated viral infections.

💡 Clinical Features of Pharyngitis and Tonsillitis

▼ Pharyngitis commonly produces sore throat, pharyngeal pain, and painful swallowing.

🎯 On examination, the affected mucosa usually appears red (erythematous) and swollen because of inflammation.

▼ Tonsillitis produces similar inflammatory changes in the tonsils. The tonsils may become enlarged, erythematous, and sometimes covered with white or yellowish exudate.

★ Severe bacterial pharyngitis may be associated with fever, tender cervical lymph nodes, and tonsillar exudate, particularly in Group A Streptococcus infection.

✦ Etiologic Agents

- ▼ Viruses are responsible for the vast majority of URTIs, accounting for more than 80% of infections.

🎯 Important viral causes include rhinoviruses, adenoviruses, coronaviruses, influenza viruses, parainfluenza viruses, and respiratory syncytial virus (RSV).

- ▼ Viral URTIs are generally self-limiting and usually require supportive treatment rather than antibiotics.

✦ Major Bacterial Causes

- ▼ *Streptococcus pyogenes* (Group A *Streptococcus*): It is the most common bacterial cause of acute pharyngitis. Infection typically produces sore throat, fever, pharyngeal redness, and tonsillar inflammation, sometimes with exudate.

🎯 An important reason to recognize Group A *Streptococcus* is its potential complications, including acute rheumatic fever and peritonsillar abscess.

- ▼ *Corynebacterium diphtheriae*: Causes diphtheria, an important bacterial respiratory infection characterized by production of a potent diphtheria toxin. Severe infection may produce a characteristic grayish-white pseudomembrane in the throat.

- ▼ *Neisseria gonorrhoeae*: Can infect the pharynx, usually following oral–genital sexual contact. Pharyngeal infection may be asymptomatic or produce sore throat and inflammation.

✦ Rare Bacterial Causes

- ▼ *Corynebacterium ulcerans* & *Arcanobacterium haemolyticum* → Can cause acute pharyngitis, especially in young people.
- ▼ *Francisella tularensis* → May cause oropharyngeal tularemia through contaminated food or water.
- ▼ Group B, C & G streptococci → Can cause pharyngitis, but Group A *Streptococcus* is the most common bacterial cause.

Major Viral Causes of Upper Respiratory Disease

✦ Rhinoviruses — Picornaviridae

- ▼ Rhinoviruses are the most common cause of the common cold and mainly infect the upper respiratory tract, especially the nasal epithelium.

🎯 Structure: They are 28–30 nm, non-enveloped, icosahedral viruses containing a positive-sense ssRNA genome.

- ▼ Properties: Rhinoviruses are acid-labile, relatively thermostable, and can survive for several days on surfaces. They replicate mainly in the nasal epithelium and grow best at about 33°C, which corresponds to the temperature of the nasopharynx.

🌟 Diversity: More than 100 serotypes exist, making repeated rhinovirus infections common.

✦ Adenoviruses — Adenoviridae

▼ Adenoviruses are 70–90 nm, non-enveloped, icosahedral viruses containing a double-stranded DNA genome. Their capsid consists of approximately 252 capsomeres.

🎯 Classification: The family includes Aviadenovirus, mainly infecting birds, and Mastadenovirus, infecting mammals including humans.

▼ General Properties: Adenoviruses are relatively resistant to environmental conditions, including low pH, bile salts, and proteolytic enzymes. This resistance allows some types to survive and replicate in the gastrointestinal tract.

✓ Pathogenesis: Adenoviruses can infect the respiratory tract, eyes, gastrointestinal tract, bladder, and liver. Some Group C adenoviruses may establish latent infections in the adenoids and tonsils.

🌟 Key Point: Adenoviruses can produce both respiratory and non-respiratory infections because of their broad tissue tropism.

💡 Coronaviruses — Coronaviridae

▼ Coronaviruses are enveloped RNA viruses, approximately 80–220 nm in size. Their characteristic “crown-like” appearance is produced by club-shaped surface peplomers.

🎯 Important Glycoproteins:

▼ S (Spike) protein: Responsible for attachment and fusion with host cells.

▼ M (Membrane) protein: Important in viral assembly and budding.

▼ HE (Hemagglutinin): Present in some coronaviruses and has acetylcholinesterase activity.

▼ Classification: Human and animal coronaviruses are mainly included within the genus Coronavirus, while Torovirus mainly infects animals and can cause gastrointestinal disease.

✓ Pathogenesis: Coronaviruses are generally species-specific and show tropism for respiratory and gastrointestinal epithelial cells. Respiratory infection can cause patchy destruction of ciliated epithelial cells, impairing normal mucociliary clearance.

✦ Parainfluenzaviruses — Paramyxoviridae

▼ Parainfluenzaviruses are enveloped, non-segmented, negative-sense ssRNA viruses. They are relatively antigenically stable, meaning major antigenic changes are uncommon.

🎯 Important Types:

▼ Type 1: Important cause of croup.

▼ Type 2: Also commonly causes croup.

▼ Type 3: More strongly associated with bronchiolitis and pneumonia.

▼ Type 4: Usually produces a milder upper respiratory infection.

▼ Pathogenesis: These viruses replicate in the respiratory epithelium, producing inflammation and swelling. Types 1 and 2 are especially associated with acute laryngotracheobronchitis (croup).

★ Key Clinical Feature: Croup typically produces a barking or seal-like cough, hoarseness, and inspiratory stridor due to upper-airway narrowing.

✦ Influenza Viruses — Orthomyxoviridae

▼ Influenza viruses are enveloped, spherical or pleomorphic viruses measuring approximately 80–120 nm. They contain a segmented negative-sense ssRNA genome.

🎯 Genome:

- ▼ Influenza A: 8 RNA segments.
- ▼ Influenza B: 8 RNA segments.
- ▼ Influenza C: 7 RNA segments.

✦ Important Viral Proteins

- ▼ Hemagglutinin (HA): Allows the virus to attach to host-cell receptors and contributes to entry.
- ▼ Neuraminidase (NA): Helps newly formed viral particles release from infected cells and facilitates spread through respiratory mucus.
- ▼ Matrix proteins (M1, M2): Important for viral structure, assembly, and release.

▼ Nucleoprotein (NP): Associates with the viral RNA and helps form the viral ribonucleoprotein complex.

▼ PB1, PB2, and PA: Form the RNA polymerase complex required for viral RNA replication and transcription.

✦ Antigenic Variation of Influenza

▼ Antigenic Drift occurs through small point mutations that gradually alter viral antigens. It is responsible for the continuous emergence of new seasonal influenza strains.

🌀 Antigenic Shift is a major antigenic change caused by reassortment of RNA segments between different influenza viruses. It can produce completely new subtypes and may lead to influenza pandemics.

🌟 Important: Antigenic shift occurs only in Influenza A.

✦ Influenza Classification and Pathogenesis

▼ Influenza viruses are classified into Types A, B, and C based mainly on their nucleoprotein and matrix proteins.

✓ Influenza A viruses are further classified according to their HA and NA surface proteins, such as H1N1 and H3N2.

▼ During infection, neuraminidase decreases mucus viscosity, helping viral particles spread through the respiratory tract. Influenza also causes destruction of the superficial respiratory mucosa.

🎯 This epithelial damage weakens normal respiratory defenses and increases susceptibility to secondary bacterial infections, particularly by *Streptococcus pneumoniae*, *Staphylococcus aureus*, and *Haemophilus influenzae*.

💡 Respiratory Syncytial Virus (RSV) — Paramyxoviridae

▼ RSV is an enveloped, non-segmented, negative-sense ssRNA virus and is an important cause of respiratory infections, especially in infants and young children.

🎯 Pathogenesis: RSV initially replicates in the nasopharyngeal epithelium and may spread downward into the lower respiratory tract, where it can cause bronchiolitis and pneumonia.

▼ Viremia is generally not detected, meaning the infection remains mainly within the respiratory tract. The immune response is important for controlling and eventually resolving the infection.

🌟 Key Point

▼ Rhinovirus → Common cold → Non-enveloped + ssRNA + >100 serotypes

▼ Adenovirus → Respiratory/eye/GI infections → Non-enveloped + dsDNA

▼ Coronavirus → Crown-like appearance → Enveloped RNA virus → Ciliated-cell damage

▼ Parainfluenza → Croup → Barking cough + stridor

▼ Influenza → HA + NA → Antigenic drift and shift → Secondary bacterial infection

▼ RSV → Nasopharynx → Lower respiratory tract → Bronchiolitis and pneumonia

Major Bacterial Causes of Respiratory Disease

💡 Streptococcus pyogenes — Group A Streptococcus

▼ Streptococcus pyogenes is a Gram-positive coccus that typically occurs in pairs or chains. It is catalase-negative and a facultative anaerobe. It is an important cause of bacterial pharyngitis and other respiratory infections.

🎯 Adherence Factors

▼ Lipoteichoic acid produces an initial weak attachment to respiratory epithelial cells.

▼ M protein and F protein help the bacteria establish stronger and more stable attachment to host cells.

💡 Immune Evasion

▼ The M protein binds Factor H, promoting breakdown of C3b and reducing phagocytosis.

▼ M protein can also bind fibrinogen, interfering with complement activity and helping the organism avoid immune destruction.

- ▼ C5a peptidase destroys C5a, reducing recruitment of neutrophils to the site of infection.

💡 Toxins and Enzymes

- ▼ Pyrogenic exotoxins (SpeA, SpeB, SpeC): Act as superantigens and are associated with severe systemic effects, including toxic shock, while some are involved in the scarlet fever rash.
- ▼ Streptolysin S: An oxygen-stable hemolysin that damages and lyses blood cells.
- ▼ Streptolysin O: An oxygen-labile hemolysin that also damages blood cells. Antibodies against it can be detected by the ASO (antistreptolysin O) test.
- ▼ Streptokinase: Breaks down blood clots, helping the bacteria spread through tissues.
- ▼ DNases A–D: Break down DNA in pus, thereby reducing its viscosity and facilitating bacterial spread.

🌟 Key Point:

S. pyogenes → M protein + C5a peptidase → immune evasion; Streptolysins + Spe toxins + Streptokinase + DNases → tissue damage and spread.

💡 *Corynebacterium diphtheriae*

- ▼ *Corynebacterium diphtheriae* is a Gram-positive, pleomorphic bacillus characterized by metachromatic granules. Its major pathogenic effect is produced by the diphtheria toxin.

🎯 Diphtheria Toxin

- ▼ The toxin is an A–B exotoxin with two functional components.
- ▼ B subunit: Binds to the heparin-binding EGF receptor on host cells and allows the toxin to enter.
- ▼ A subunit: Inactivates elongation factor 2 (EF-2), stopping protein synthesis and ultimately causing cell death.

🌟 Key Point:

C. diphtheriae → Diphtheria toxin → EF-2 inactivation → inhibition of protein synthesis.

💡 *Neisseria gonorrhoeae*

▼ *Neisseria gonorrhoeae* is a Gram-negative diplococcus and is characteristically oxidase-positive. It can colonize mucosal surfaces and uses several factors to attach to and invade host cells.

🎯 Pathogenesis

- ▼ Pili: Responsible for initial attachment to epithelial cells and help the organism establish infection.
- ▼ Opa protein: Promotes tight association with host cells and facilitates bacterial interaction and migration through tissues.
- ▼ Por protein: Helps the organism evade immune destruction by inhibiting phagolysosome fusion.

▼ LOS (lipooligosaccharide): Acts as an important inflammatory component, stimulating inflammation and tissue damage.

🌟 Key Point

N. gonorrhoeae → Pili for attachment → Opa for tight association → Por for immune evasion → LOS for inflammation.

General Management of URTI

💡 Diagnosis

▼ Diagnosis of Upper Respiratory Tract Infections (URTIs) depends on the suspected causative organism and clinical presentation. A throat or respiratory swab may be collected for laboratory testing.

🎯 Streptococcus pyogenes

▼ A throat culture can be used to detect S. pyogenes. Rapid direct antigen tests are also available and provide faster results, although they may have lower sensitivity than culture.

🎯 Corynebacterium diphtheriae

▼ If diphtheria is suspected, specific culture and laboratory testing should be performed, especially when a characteristic pseudomembrane is present.

Neisseria gonorrhoeae

▼ When *N. gonorrhoeae* infection is suspected, appropriate specific microbiological testing should be performed using samples from the affected site.

Treatment

▼ Viral URTI: Most viral infections are self-limiting, so treatment is mainly supportive and symptomatic, such as fluids, rest, and relief of fever or pain. Antibiotics are not effective against viruses.

Streptococcus pyogenes:

▼ Penicillin is the preferred treatment. In patients with certain penicillin allergies, alternatives such as erythromycin or appropriate cephalosporins may be used depending on the type of allergy and clinical situation.

Corynebacterium diphtheriae:

▼ Treatment requires Diphtheria Antitoxin to neutralize circulating toxin, together with an appropriate antibiotic such as penicillin or erythromycin to eliminate the organism and reduce transmission.

Neisseria gonorrhoeae:

▼ Ceftriaxone is the recommended treatment in current guidelines. Older regimens included drugs such as ciprofloxacin only when susceptibility is confirmed, because resistance to fluoroquinolones is common.

🌟 Key Point

▼ Viral URTI → Supportive treatment

▼ *S. pyogenes* → Penicillin

▼ *C. diphtheriae* → Antitoxin + antibiotic

▼ *N. gonorrhoeae* → Ceftriaxone

Middle Respiratory Tract Infections (MRTI)

💡 Anatomical Sites

▼ Middle Respiratory Tract Infections (MRTIs) involve the epiglottis, aryepiglottic tissues, larynx, trachea, and bronchi. Infection and inflammation in these areas can cause airway narrowing, cough, and difficulty breathing.

💡 Epiglottitis

▼ Epiglottitis is an acute inflammation of the epiglottis and surrounding tissues. It may develop suddenly and can rapidly become serious because swelling can obstruct the airway.

🎯 Clinical Features: Abrupt throat or neck pain, fever, difficulty swallowing, and inspiratory stridor are common. The major danger is acute airway obstruction.

★ Key Point: Epiglottitis is different from most other MRTIs because it is often bacterial rather than viral.

💡 Laryngitis / Croup

▼ Laryngitis and croup involve inflammation of the larynx, including the vocal cords and subglottic structures. Croup occurs mainly in young children and is usually viral.

🎯 Clinical Features: The characteristic findings are a barking or seal-like cough, hoarse phonation, and sometimes inspiratory stridor due to narrowing of the upper airway.

▼ About 90% of croup and laryngitis cases are viral, with parainfluenza viruses being important causes.

💡 Bronchitis / Tracheobronchitis

▼ Bronchitis and tracheobronchitis involve inflammation of the bronchi and trachea. They commonly produce a persistent cough with increasing airway secretions.

🎯 Clinical Features: The cough may initially produce clear sputum, which can later become purulent as inflammation progresses. Wheezing or chest discomfort may also occur.

✦ Etiologic Agents

- ▼ Most MRTIs are viral, particularly croup and laryngitis, where approximately 90% of cases are caused by viruses.
- ▼ Epiglottitis is an important exception, as it is often bacterial and can cause rapid airway obstruction.

✦ Key Point

- ▼ Epiglottitis → Fever + throat/neck pain + stridor → Airway obstruction risk
- ▼ Croup/Laryngitis → Barking cough + hoarseness → Mostly viral
- ▼ Bronchitis/Tracheobronchitis → Cough + sputum → Mostly viral

Bacterial Pathogens of the Middle and Lower Respiratory Tract

✦ Haemophilus influenzae

- ▼ Haemophilus influenzae is a small Gram-negative rod that can cause infections of the middle and lower respiratory tract, including epiglottitis, bronchitis, and pneumonia.

🎯 Physiology

▼ It requires two important growth factors: X factor (hemin) and V factor (NAD). These requirements are useful for its laboratory identification.

✦ Structure

▼ Its outer membrane contains LPS, and some strains possess a polysaccharide capsule. Encapsulated strains are classified into six serotypes (a–f).

✦ Serotype b (Hib) has a PRP (polyribosylribitol phosphate) capsule and is the most invasive serotype because the capsule strongly protects the organism from phagocytosis.

🎯 Pathogenesis

▼ Pili and adhesins help the bacteria attach to and colonize respiratory epithelial cells.

▼ The PRP capsule is the major virulence factor because it is antiphagocytic, allowing the bacteria to survive and spread in the host.

▼ The organism also produces IgA1 protease, which breaks down secretory IgA and helps it colonize the respiratory mucosa.

✦ Streptococcus pneumoniae

▼ Streptococcus pneumoniae is a Gram-positive coccus, usually seen in pairs or short chains. It is catalase-negative and a facultative anaerobe and is an important cause of respiratory infections and pneumonia.

🎯 Colonization and Migration

▼ Pneumococci attach to respiratory epithelial cells using surface protein adhesins.

▼ Secretory IgA protease breaks down IgA in respiratory secretions, helping the bacteria avoid being trapped and removed by mucociliary clearance.

💡 Tissue Destruction

▼ Pneumolysin is an important toxin that forms pores in host-cell membranes, causing cellular injury.

▼ Teichoic acid and peptidoglycan activate the immune system and stimulate the release of inflammatory cytokines such as IL-1 and TNF- α , contributing to inflammation and tissue damage.

💡 Survival from Phagocytosis

▼ The polysaccharide capsule is the major antiphagocytic factor, protecting pneumococci from being engulfed by immune cells.

▼ Pneumolysin can also interfere with the oxidative burst of phagocytes, helping the bacteria survive inside the host.

🌟 Key Point

▼ *H. influenzae* → X + V factors → PRP capsule → Antiphagocytic → IgA1 protease

▼ *S. pneumoniae* → Gram-positive diplococcus → Capsule + IgA protease → Pneumolysin → Respiratory tissue damage

Pathogenesis of Neisseria meningitidis

Nasopharyngeal colonization



 Pili-mediated attachment



 Invasion/transcytosis through epithelium



 Bacteremia



 Capsule prevents phagocytosis



LOS-mediated systemic inflammation



 Meningitis ± meningococemia

 Clinical Disease

 Acute bacterial meningitis

 Meningococemia

 Fever + severe systemic toxicity

 Petechial/purpuric rash 

 DIC

 Septic shock

 Waterhouse–Friderichsen syndrome → adrenal hemorrhage associated with fulminant meningococemia

Respiratory Connection

 *N. meningitidis* primarily causes meningitis and meningococemia, rather than being a major routine cause of epiglottitis.

 Exam pearl:

If an MCQ asks for a classic bacterial cause of acute epiglottitis, think *H. influenzae* type b, although *S. aureus*, streptococci and other organisms can also cause it.

 Prevention

 Meningococcal vaccination is the key preventive strategy.

 Close contacts of invasive meningococcal disease require chemoprophylaxis, commonly with agents such as rifampin, ciprofloxacin, or ceftriaxone, depending on age, pregnancy, resistance patterns and local recommendations.

④ ☠ Superantigen

TSST-1(=Toxic shock syndrome toxin-1)



Massive, nonspecific T-cell activation



 Cytokine release



 Fever + hypotension + rash



 Multisystem involvement

→ Toxic shock syndrome

5 Important enzymes

Enzyme Major function

 Coagulase Fibrin formation

 Hyaluronidase Tissue spread

 Staphylokinase Fibrinolysis

 Penicillinase Hydrolyzes penicillin

 Lipases Help survival/spread in tissues

 Respiratory Disease

S. aureus can cause:

 Pneumonia

 Lung abscess

 Empyema

 Necrotizing pneumonia

 /  Epiglottitis — possible, but not the classic organism

 Secondary bacterial pneumonia following viral respiratory infection, classically after influenza.

 Classic Exam Connection

 Influenza → damaged respiratory epithelium → secondary *S. aureus* pneumonia

 Severe *S. aureus* pneumonia may be necrotizing, especially with toxin-producing strains.

  Adenylate cyclase toxin

▼ Directly increases intracellular cAMP.



● Impairs neutrophil/macrophage function



● Reduces phagocytosis and killing

3 🧬 Tracheal cytotoxin

▼ ▼ Targets ciliated epithelial cells.



💥 Ciliary damage



🚫 Impaired mucociliary clearance



🫁 Accumulation of mucus/secretions



😷 Severe paroxysmal cough

★ This is a major explanation for the characteristic cough.

📊 CLINICAL COURSE — 3 CLASSIC STAGES

1 ☁️ CATARRHAL STAGE

 ~1–2 weeks

✓Symptoms resemble an ordinary URI:

 Runny nose

 Nasal congestion

 Mild cough

 Low-grade or absent fever

 Watery eyes

 MOST CONTAGIOUS STAGE

 This is why pertussis can spread before the classic “whoop” develops.

2 PAROXYSMAL STAGE

 ~2–6 weeks or longer

Repeated violent coughing:

 Cough → cough → cough → cough



 Difficulty inhaling



 Inspiratory “WHOOOP”



 Post-tussive vomiting

Other findings:

 Exhaustion

 Cyanosis during severe attacks

 Apnea, especially in infants

 Infant warning! 

Infants may NOT produce the classic whoop.

Instead:

 Apnea + cyanosis + respiratory distress can be prominent.

3 CONVALESCENT STAGE

 Weeks → months

 Gradual reduction in coughing

 Paroxysms become less frequent

 Persistent cough may last for weeks or even months

 This prolonged cough is sometimes called the “100-day cough.”

 DIAGNOSIS of *Bordetella pertussis*

 Preferred modern approach

 Nasopharyngeal specimen → PCR Fast and sensitive.

 Culture

Traditionally cultured on:

 Bordet-Gengou medium

 Regan-Lowe medium 

 A proper nasopharyngeal swab/aspirate is important; a routine throat swab is less useful.

 CBC

A classic laboratory clue:

 Marked lymphocytosis 

Especially in children.

 TREATMENT

 Macrolides

 Azithromycin

 Clarithromycin

 Erythromycin

 Why treat?

Antibiotics:

 Reduce transmission

 Eradicate the organism

✗ May have limited effect on the established cough once toxin-mediated epithelial damage has occurred

PREVENTION

 DTaP / Tdap vaccination

 Pertussis prevention is primarily through pertussis-containing vaccines.

 DTaP → children

 Tdap → adolescents/adults and during pregnancy according to vaccination recommendations

 Post-exposure prophylaxis

Close contacts may require macrolide prophylaxis, particularly those at high risk of severe disease.

  Antibiotic Implication

 Because there is no peptidoglycan cell wall:

✗ Penicillins

✗ Cephalosporins

✗ Carbapenems

✗ Vancomycin is ineffective

 Exam pearl:

✦ No cell wall → β -lactams don't work.

✦ Macrolides, tetracyclines, or appropriate respiratory fluoroquinolones can be used depending on patient factors and local resistance patterns.

🔗 PATHOGENESIS

1 Attachment

P1 adhesin 🔗



Binds to receptors on ciliated respiratory epithelial cells



2 Ciliary damage

🦠 Attachment

→ ciliostasis 🚫

→ epithelial-cell injury/destruction

→ impaired mucociliary clearance



💧 Accumulation of secretions + inflammatory response



😷 Persistent cough

3 CARDS TOXIN

★ An important additional virulence factor is CARDS toxin (Community-Acquired Respiratory Distress Syndrome toxin).

It contributes to:

🫁 Respiratory epithelial injury

🔥 Inflammation

↑ Host inflammatory responses

4 Immune-mediated disease

M. pneumoniae can stimulate strong inflammatory responses.

🔥 Cytokine release

→ TNF- α

→ IL-1

→ IL-6

→ tissue inflammation

✚ Some manifestations of infection are related to immune-mediated mechanisms, not simply direct bacterial destruction.

CLINICAL FEATURES

★ Classic presentation:

Gradual onset rather than sudden severe illness.

 Low-grade fever

 Malaise

 Headache

 Persistent dry/nonproductive cough

 Mild dyspnea

 Sore throat

 Atypical pneumonia

 Classic phrase:

 “Walking pneumonia”

Patients may appear less ill than expected despite significant radiographic abnormalities.

EXTRAPULMONARY MANIFESTATIONS

 *M. pneumoniae* can produce disease outside the lungs:

 Hemolytic anemia → classically associated with cold agglutinins

 Encephalitis

 Meningitis

Myocarditis/pericarditis

 Skin rashes/erythema multiforme

 Otitis media

 HIGH-YIELD

Mycoplasma → pneumonia + cold agglutinin hemolytic anemia

 DIAGNOSIS

 NAAT/PCR

A very useful modern diagnostic method.

Serology

Historically important, particularly demonstration of an appropriate antibody response.

Culture

Possible but slow and technically difficult, so not routinely preferred.

 Correction: Don't memorize serology as the only or universally preferred modern test; NAAT/PCR is an important current diagnostic method.

CLINICAL FEATURES

Most commonly affects:

 School-aged children

 Adolescents

 Young adults

Can cause:

 Pharyngitis

 Sinusitis

 Bronchitis

 Atypical pneumonia

Often:

Mild

Gradual onset

Persistent cough

Fever may be relatively low

 Severe disease can occur, especially in vulnerable patients.

 DIAGNOSIS

 NAAT/PCR

Useful for detecting *C. pneumoniae*.

 Serology

May be used in selected settings, particularly for epidemiologic or retrospective diagnosis.

 Culture

Difficult and not routinely used.

 CLASSIC PATTERN

Symptoms frequently become worse at night.

🌙 Nighttime

→ increased symptoms

→ barking cough + stridor

🧠 DIAGNOSIS

Usually clinical.

X-ray if obtained:

🏠 “Steeple sign” ★

→ subglottic narrowing on AP neck radiograph.

📌 Imaging is not routinely necessary in typical uncomplicated croup.

📌 MANAGEMENT

● Mild

💧 Fluids/hydration

👁️ Observation

Minimize agitation

⚠ Humidified air

It may provide comfort, but routine humidified air is not a proven primary treatment for croup.

● Moderate–Severe

💊 Dexamethasone ★

Reduces:

🔥 Airway inflammation



🫁 Airway edema



😌 Improved breathing

💨 Nebulized epinephrine

Used for moderate/severe croup with significant stridor, especially at rest.

⚠ Severe cases require close observation because symptoms can recur after the effect of epinephrine wears off.

ETIOLOGY

MOSTLY VIRAL

Common causes include:

 Influenza A/B

 RSV

 Rhinovirus

 Coronavirus

 Parainfluenza

 Adenovirus

 Bacterial causes are uncommon

Important exceptions:

 *Bordetella pertussis*

 *Mycoplasma pneumoniae*

 Chlamydia pneumoniae

 Do not automatically prescribe antibiotics simply because sputum becomes yellow/green.

Purulent sputum can result from neutrophils and inflammation, not necessarily bacterial infection.

 CLINICAL FEATURES

 Main symptom:

 COUGH

It may initially be:

➡ Dry

and later become:

➡ Productive

Other findings:

 Sputum production

 Chest discomfort

 Wheezing

 Rhonchi

 Mild fever sometimes

 Fatigue

 Duration

The cough can persist for several weeks, even after the infection has resolved.

 BRONCHITIS vs PNEUMONIA ★★ ★

🔪 Acute bronchitis = cough WITHOUT evidence of pneumonia.

Other important organisms:

 Haemophilus influenzae

 Staphylococcus aureus

 Moraxella catarrhalis

 Klebsiella pneumoniae

 Pseudomonas aeruginosa → particularly healthcare-associated/high-risk patients

 Atypical organisms

 Mycoplasma pneumoniae

 Chlamydia pneumoniae

 Legionella pneumophila And numerous respiratory viruses.

 Exam pearl: Don't interpret “most common bacterial cause” as universally true for every age group or clinical setting—the likely pathogen varies with age, comorbidities, vaccination, and whether infection is community- or healthcare-associated.

B. CHRONIC PNEUMONIA

 Course

Develops over:

Weeks → months → sometimes years

Features

 Persistent fever

 Fatigue

 Weight loss

 Night sweats

 Loss of appetite

 Chronic cough

 Hemoptysis

 Important causes

 *Mycobacterium tuberculosis*

Other possibilities:

Other mycobacteria

 Fungi such as *Histoplasma*

Other chronic pulmonary infections

II. PLEURAL COMPLICATIONS

1 EMPYEMA

Definition

▼ Empyema = accumulation of pus/infected fluid in the pleural space.  

Often develops as a complication of pneumonia.

 Important organisms

 *Streptococcus pneumoniae*

 *Staphylococcus aureus*

 Streptococci

 Anaerobic bacteria

 *Pseudomonas aeruginosa* in selected healthcare-associated settings

 Clinical clues

 Persistent fever despite treatment

 Pleuritic chest pain

 Dyspnea

 Pleural effusion

 Empyema usually requires drainage in addition to appropriate antibiotics.

2 LUNG ABSCESS

 Definition

A localized cavity containing pus and necrotic lung tissue.

 Infection → tissue necrosis → cavity formation → pus

 CLASSIC CAUSE

 Aspiration of oropharyngeal contents

Especially:

 Altered consciousness

 Alcohol misuse

 Neurologic disorders

 Poor dental hygiene

 Aspiration risk

➔ Mixed anaerobic oral flora are classic causes.

 Clinical Features

 Persistent fever

 Chronic cough

 Purulent sputum

 Foul-smelling sputum → suggests anaerobic infection

 Possible hemoptysis

 Imaging

Often shows:

 Cavitory lesion

with an:

 Air-fluid level

★ Very high-yield imaging clue.

 III. ENTEROBACTERIACEAE

General Characteristics

Enterobacteriaceae are a large family of:

 Gram-negative rods

 Oxidase negative

 Catalase positive

 Glucose fermenters

 Facultative anaerobes

 Usually non-spore-forming

 MACCONKEY AGAR

MacConkey differentiates:

 Lactose fermenters

➡ Produce pink/red colonies

Examples:

E. coli

Klebsiella

Enterobacter

 Non-lactose fermenters

➡ Produce colorless/pale colonies

Examples:

Salmonella

Shigella

Proteus

 Correction: The appearance is generally described as pink/red for lactose fermenters, not specifically “pink-purple.”

 MAJOR VIRULENCE FACTORS

1 LPS / Endotoxin

LPS contains:

Lipid A ★



Cytokine release



🔥 Fever



🩸 Vasodilation + vascular leakage



💥 Hypotension



🚒 Septic shock

2 Capsule

Prevents:

🧬 Complement-mediated killing

● Phagocytosis

★ Example

Klebsiella pneumoniae has a prominent capsule.

3 Siderophores

Bacteria need iron to grow.

Siderophores:

 Bind iron



Transport iron back to bacteria

Example:

Enterobactin

4 Type III Secretion System

Think:

 “Bacterial molecular syringe”

Injects bacterial effector proteins directly into host cells.

IV. PSEUDOMONAS AERUGINOSA

IDENTIFICATION

 Gram-negative rod

 Motile

 Obligate aerobe

 Oxidase positive

 Non-lactose fermenter

 Can grow at 42°C 

CHARACTERISTIC PIGMENTS

 Pyocyanin

Produces:

Blue-green pigment

 Pyoverdine

Produces:

Yellow-green fluorescent pigment

🧠 Memory:

PyoCYANIN = CYAN → blue/blue-green

🌐 OTHER CLUES

👃 Characteristic odor

Often described as:

🍇 Grape-like / fruity odor

🧫 Growth

Can grow in relatively nutrient-poor environments and readily forms biofilms.

🫁 MAJOR RESPIRATORY ASSOCIATIONS

★ Cystic fibrosis

In CF:

 Thick mucus



 Colonization



 Biofilm formation



 Mucoid Pseudomonas

 Alginate

Produces an abundant alginate-rich extracellular matrix:

 →  Mucoid biofilm

This helps protect bacteria from:

 Phagocytosis

 Antibiotics

 Host immune responses

 MAJOR TOXINS

1 Exotoxin A

Mechanism:

ADP-ribosylates EF-2



 Protein synthesis

This mechanism is conceptually similar to diphtheria toxin.

 Memory:

▼ Pseudomonas + Diphtheria → EF-2 inhibition

2 Elastases

LasA + LasB

Break down:

 Elastin

 Tissue components

 Other host proteins

➡ Contribute to pulmonary tissue destruction.

ANTIBIOTIC RESISTANCE

P. aeruginosa is notorious for multidrug resistance.

Mechanisms include:

 Low outer-membrane permeability

 β-lactamases

 Efflux pumps

 Biofilm formation

 Target modification

 Therefore, treatment must be guided by local susceptibility patterns and culture results when appropriate.

HIGH-YIELD PSEUDOMONAS ASSOCIATIONS

Think:

“PSEUDO = BLUE-GREEN + CF + BURNS + HOSPITAL”

 Pyocyanin

 Pyoverdine

 Cystic fibrosis

 Burn/wound infections

 Healthcare-associated infections

 Ventilator-associated pneumonia

 Bacteremia

Typical granuloma:

 Central caseous necrosis



 Epithelioid macrophages



 Langhans giant cells



 Lymphocytic rim

 Key concept

Granuloma = containment, NOT necessarily eradication.

The organism can remain dormant.

 PRIMARY vs SECONDARY TB

1 PRIMARY TB

Occurs after initial infection.

Usually:

 Lower/middle lung regions or subpleural focus+  Hilar lymph-node involvement

The combination is called the:

★ Ghon complex

Ghon focus + involved regional lymph nodes.

2 SECONDARY TB

Usually represents:

🔄 Reactivation of latent TB

Classically favors:

↑ Apices of the upper lobes

Why?

💡 Higher oxygen tension

Because M. tuberculosis is an obligate aerobe.

🔥 Reactivation may cause:

🫁 Cavitation

😷 Chronic cough

🌙 Night sweats

⚖️ Weight loss

🌡️ Fever

🩸 Hemoptysis

DIAGNOSIS

1. Ziehl–Neelsen stain

Acid-fast bacilli appear:

 Red/pink rods

against:

 Blue background

2. Auramine–Rhodamine

Fluorescent staining:

 Acid-fast bacilli fluoresce.

Often more sensitive for screening microscopy.

3. Molecular testing

 Xpert MTB/RIF / Xpert MTB/RIF Ultra

Can rapidly detect:

 M. tuberculosis complex

+  Rifampicin resistance

This is extremely important for rapid diagnosis and resistance assessment.

4. Culture

Culture remains important because it allows:

 Confirmation

 Drug susceptibility testing

But:

 Culture is slow because *M. tuberculosis* grows very slowly.

 LATENT TB TESTING

Important distinction:

 IGRA

Interferon-gamma release assay

 Tuberculin skin test

Mantoux/TST

These detect TB infection, but they do not by themselves prove active pulmonary TB.

★ Very important exam distinction.

 TB TREATMENT — RIPE

 Initial intensive phase

Usually:

R — Rifampicin 

I — Isoniazid 

P — Pyrazinamide 

E — Ethambutol 

➔ RIPE

For drug-susceptible pulmonary TB, a common standard regimen is:

2 months RIPE → 4 months continuation therapy

= approximately 6 months total, although treatment duration can vary according to disease site, resistance, regimen, and clinical circumstances.

✚ Correction: Don't memorize "6–9 months" as the universal standard. Six months is the classic standard regimen for drug-susceptible pulmonary TB, with longer courses used in selected situations.